

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection

A city surveillance system must operate under the following conditions:

- If latency > 100 ms → system fails
- If accuracy < 85% → unacceptable
- Hardware supports only moderate computation Models available:

Viola-Jones: 78%, low latency

YOLO: 92%, moderate latency

Deep Learning: 95%, high latency

Answer:

Model	Accuracy	Latency	Computation	Result
Viola-Jones	78%	Low	Low	✗ Accuracy too low
YOLO	92%	Moderate	Moderate	☑ Satisfies all
Deep Learning	95%	High	High	✗ Not suitable for hardware

Evaluation:

Viola-Jones is fast but its accuracy is only 78%, which is below the required 85%. Deep Learning provides 95% accuracy but requires high computation, which is unsuitable for moderate hardware.

Selected Solution: YOLO

YOLO provides **92% accuracy**, moderate latency, and moderate computational requirements. Therefore, it satisfies all the given constraints and provides a good balance between accuracy and resource requirements.

Final_Decision:

YOLO is the most practical solution for city surveillance because it provides sufficient accuracy while remaining suitable for the available hardware

2. Autonomous Vehicle Detection

LogicA vehicle detection system must:

- Prioritize safety (accuracy > 90%)
- Maintain response time < 50 ms Options:
Method A: 88% accuracy, 30 ms
Method B: 93% accuracy, 70 ms
Method C: 91% accuracy, 45 ms

Answer:

Method	Accuracy	Response Time	Result
A	88%	30 ms	✗ Accuracy < 90%
B	93%	70 ms	✗ Time > 50 ms
C	91%	45 ms	☑ Satisfies both

Evaluation:

Method A is fast but fails the accuracy requirement. Method B has high accuracy but its response time is too slow for an autonomous vehicle. Method C provides 91% accuracy and responds within 45 ms.

Selected Solution: Method C

Method C satisfies **both safety constraints**: accuracy greater than 90% and response time below 50 ms.

Final_Decision:

Method C should be selected because it provides the required accuracy and real-time response, making it practical for autonomous vehicle detection.

3. Motion Analysis Decision ProblemA

traffic system requires:

- Accurate motion direction detection
 - Robustness in dense traffic
- Results:
- Optical Flow: accurate but noisy in dense scenes
- Motion Estimation: stable but less detailed

Answer:

Approach	Accuracy/Detail	Dense Traffic Robustness	Result
Optical Flow	High	Low	✗ Noisy in crowds
Motion Estimation	Moderate	High	☑ Suitable

Evaluation:

Optical Flow can provide detailed and accurate motion information, but it becomes noisy when many vehicles or objects are present. In a dense traffic environment, this noise can reduce reliability.

Motion Estimation is less detailed but provides more stable results in crowded scenes. Since robustness is an important requirement, stability is more valuable than highly detailed motion information.

Selected Solution: Motion Estimation

Final Decision:

Motion Estimation is more suitable for high-density traffic because it provides **stable and reliable motion detection**, even though it gives less detail.

4. Background Modeling Evaluation

GMM-based system produces:

- Static scenes: 90% accuracy
- Dynamic scenes: 65% accuracy

Requirement:

System must maintain $\geq 80\%$ accuracy across all scenarios

Answer:

Scenario	GMM Accuracy	Required Accuracy	Result
Static scenes	90%	$\geq 80\%$	☑ Pass
Dynamic scenes	65%	$\geq 80\%$	✗ Fail

Evaluation:

GMM performs well in static scenes with 90% accuracy. However, its accuracy decreases significantly to 65% in dynamic scenes. Since the system must maintain at least 80% accuracy **in all scenarios**, GMM alone does not satisfy the requirement.

Decision:

GMM should **not be used as the sole solution**. It should be improved or combined with another technique that can handle dynamic backgrounds more effectively.

Final Decision:

A **hybrid or enhanced background-modeling approach** should be considered to improve performance in dynamic scenes. If no single method satisfies all constraints, combining techniques is a practical solution.

5. Depth Estimation**Strategy Constraints:**

- Only one camera available
 - Moderate depth accuracy acceptable
- Options:
- Stereo Vision: high accuracy, needs two cameras
- Structure from Motion: moderate accuracy, single camera

Answer:

Method	Cameras Required	Accuracy	Constraint Satisfaction
Stereo Vision	2	High	✗ Fails camera constraint
Structure from Motion	1	Moderate	✓ Satisfies all

Evaluation:

Stereo Vision can provide high-quality depth information, but it requires two cameras. Since the system has only one camera, Stereo Vision cannot be directly used. Structure from Motion can estimate depth using a **single camera** and provides moderate accuracy, which is sufficient according to the requirement.

Selected Solution: Structure from Motion**Final Decision:**

Structure from Motion is the most feasible solution because it works with the available **single-camera setup** and provides the required moderate depth accuracy. It also avoids the additional hardware cost of using a second camera.

Overall Conclusion

Among the five problems, the selected solutions are:

1. **Smart Surveillance** → **YOLO**
2. **Autonomous Vehicle Detection** → **Method C**
3. **Motion Analysis** → **Motion Estimation**
4. **Background Modeling** → **Hybrid/Enhanced approach**
5. **Depth Estimation** → **Structure from Motion**

Code of Conduct:

I V G ABBIRAMY – 192524097 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V G ABBIRAMY

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution

Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough	Good	Basic	Poor or no

		analysis with validated results and performance evaluation	analysis with reasonable validation	analysis with limited validation	analysis
Professionalism	5%	Highly professional, well documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation